AWS CLI & Ekctl installed

```
ubuntu@ubuntu:~$ aws --version
aws-cli/2.33.12 Python/3.13.11 Linux/6.8.0-101-generic exe/x86_64.ubuntu.24
ubuntu@ubuntu:~$ curl --location "https://github.com/weaveworks/eksctl/releases/latest/download/eksctl_Linux_amd64.tar.gz" -o eksctl.tar.gz

tar -xzf eksctl.tar.gz
sudo mv eksctl /usr/local/bin
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0
100 35.4M  100 35.4M    0     0   152k      0  0:03:58  0:03:58 --:--:-- 5035k
```

```
ubuntu@ubuntu:~$ eksctl version
0.224.0
ubuntu@ubuntu:~$
```

Kubectl installed

```
ubuntu@ubuntu:~$ sudo su -
root@ubuntu:~# kubectl version --client
Client Version: v1.30.0
Kustomize Version: v5.0.4-0.20230601165947-6ce0bf390ce3
root@ubuntu:~#
```

AWS Configured

```
root@ubuntu:~# aws sts get-caller-identity
{
    "UserId": "AIDARIBG2TVVKB[redacted]",
    "Account": "0859807[redacted]0",
    "Arn": "arn:aws:iam::0859807[redacted]"
}
root@ubuntu:~#
```

EKS Cluster created

```
root@ubuntu:~# eksctl create cluster \
--name demo-eks \
--region us-east-2 \
--nodegroup-name demo-nodes \
--node-type t3.small \
--nodes 1
2026-03-15 13:59:37 [ℹ]  eksctl version 0.224.0
2026-03-15 13:59:37 [ℹ]  using region us-east-2
2026-03-15 13:59:39 [ℹ]  setting availability zones to [us-east-2b us-east-2c us-east-2a]
2026-03-15 13:59:39 [ℹ]  subnets for us-east-2b - public:192.168.0.0/19 private:192.168.96.0/19
2026-03-15 13:59:39 [ℹ]  subnets for us-east-2c - public:192.168.32.0/19 private:192.168.128.0/19
2026-03-15 13:59:39 [ℹ]  subnets for us-east-2a - public:192.168.64.0/19 private:192.168.160.0/19
2026-03-15 13:59:39 [ℹ]  nodegroup "demo-nodes" will use "" [AmazonLinux2023/1.34]
2026-03-15 13:59:39 [!]  Auto Mode will be enabled by default in an upcoming release of eksctl. This means managed node groups and managed networking add-ons
will no longer be created by default. To maintain current behavior, explicitly set 'autoModeConfig.enabled: false' in your cluster configuration. Learn more:
https://eksctl.io/usage/auto-mode/
2026-03-15 13:59:39 [ℹ]  using Kubernetes version 1.34
2026-03-15 13:59:39 [ℹ]  creating EKS cluster "demo-eks" in "us-east-2" region with managed nodes
2026-03-15 13:59:40 [ℹ]  will create 2 separate CloudFormation stacks for cluster itself and the initial managed nodegroup
2026-03-15 13:59:40 [ℹ]  if you encounter any issues, check CloudFormation console or try 'eksctl utils describe-stacks --region=us-east-2 --cluster=demo-eks'
2026-03-15 13:59:40 [ℹ]  Kubernetes API endpoint access will use default of {publicAccess=true, privateAccess=false} for cluster "demo-eks" in "us-east-2"
2026-03-15 13:59:40 [ℹ]  CloudWatch logging will not be enabled for cluster "demo-eks" in "us-east-2"
2026-03-15 13:59:40 [ℹ]  you can enable it with 'eksctl utils update-cluster-logging --enable-types={SPECIFY-YOUR-LOG-TYPES-HERE (e.g. all)} --region=us-east-
2 --cluster=demo-eks'
2026-03-15 13:59:40 [ℹ]  default addons metrics-server, vpc-cni, kube-proxy, coredns were not specified, will install them as EKS addons
2026-03-15 13:59:40 [ℹ]
2 sequential tasks: { create cluster control plane "demo-eks",
    2 sequential sub-tasks: {
        2 sequential sub-tasks: {
            1 task: { create addons },
            wait for control plane to become ready,
        },
        create managed nodegroup "demo-nodes",
    }
}
2026-03-15 13:59:40 [ℹ]  building cluster stack "eksctl-demo-eks-cluster"
2026-03-15 13:59:42 [ℹ]  deploying stack "eksctl-demo-eks-cluster"
2026-03-15 14:00:12 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-cluster"
2026-03-15 14:00:44 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-cluster"
2026-03-15 14:01:46 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-cluster"
```

```
2026-03-15 13:59:42 [ℹ]  deploying stack "eksctl-demo-eks-cluster"
2026-03-15 14:00:12 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-cluster"
2026-03-15 14:00:44 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-cluster"
2026-03-15 14:01:46 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-cluster"
2026-03-15 14:02:49 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-cluster"
2026-03-15 14:03:51 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-cluster"
2026-03-15 14:04:53 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-cluster"
2026-03-15 14:05:55 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-cluster"
2026-03-15 14:06:57 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-cluster"
2026-03-15 14:08:00 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-cluster"
2026-03-15 14:08:09 [!]  recommended policies were found for "vpc-cni" addon, but since OIDC is disabled on the cluster, eksctl cannot configure the requeste
 permissions; the recommended way to provide IAM permissions for "vpc-cni" addon is via pod identity associations; after addon creation is completed, add all
recommended policies to the config file, under `addon.PodIdentityAssociations`, and run `eksctl update addon`
2026-03-15 14:08:09 [ℹ]  creating addon: vpc-cni
2026-03-15 14:08:10 [ℹ]  successfully created addon: vpc-cni
2026-03-15 14:08:11 [ℹ]  creating addon: kube-proxy
2026-03-15 14:08:11 [ℹ]  successfully created addon: kube-proxy
2026-03-15 14:08:12 [ℹ]  creating addon: coredns
2026-03-15 14:08:13 [ℹ]  successfully created addon: coredns
2026-03-15 14:10:21 [ℹ]  building managed nodegroup stack "eksctl-demo-eks-nodegroup-demo-nodes"
2026-03-15 14:10:24 [ℹ]  deploying stack "eksctl-demo-eks-nodegroup-demo-nodes"
2026-03-15 14:10:25 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-nodegroup-demo-nodes"
2026-03-15 14:10:57 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-nodegroup-demo-nodes"
2026-03-15 14:11:33 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-nodegroup-demo-nodes"
2026-03-15 14:13:23 [ℹ]  waiting for CloudFormation stack "eksctl-demo-eks-nodegroup-demo-nodes"
2026-03-15 14:13:23 [ℹ]  waiting for the control plane to become ready
2026-03-15 14:13:30 [✔]  saved kubeconfig as "/root/.kube/config"
2026-03-15 14:13:30 [ℹ]  no tasks
2026-03-15 14:13:30 [✔]  all EKS cluster resources for "demo-eks" have been created
2026-03-15 14:13:33 [ℹ]  nodegroup "demo-nodes" has 1 node(s)
2026-03-15 14:13:33 [ℹ]  node "ip-192-168-66-214.us-east-2.compute.internal" is ready
2026-03-15 14:13:33 [ℹ]  waiting for at least 1 node(s) to become ready in "demo-nodes"
2026-03-15 14:13:33 [ℹ]  nodegroup "demo-nodes" has 1 node(s)
2026-03-15 14:13:33 [ℹ]  node "ip-192-168-66-214.us-east-2.compute.internal" is ready
2026-03-15 14:13:33 [✔]  created 1 managed nodegroup(s) in cluster "demo-eks"
2026-03-15 14:13:36 [ℹ]  creating addon: metrics-server
2026-03-15 14:13:36 [ℹ]  successfully created addon: metrics-server
2026-03-15 14:13:47 [ℹ]  kubectl command should work with "/root/.kube/config", try 'kubectl get nodes'
2026-03-15 14:13:47 [✔]  EKS cluster "demo-eks" in "us-east-2" region is ready
root@ubuntu:~#
```

```
2026-03-15 14:13:47 [✔]  EKS cluster "demo-eks" in "us-east-2" region is ready
root@ubuntu:~# kubectl get nodes
NAME                                           STATUS   ROLES    AGE     VERSION
ip-192-168-66-214.us-east-2.compute.internal   Ready    <none>   3m20s   v1.34.4-eks-f69f56f
root@ubuntu:~#
```

ubuntu | 9% | 0.66 GB / 3.78 GB | 0.01 Mb/s | 0.00 Mb/s | 21 hours | abi (x3) | /: 50% | /boot: 13%

CloudFormation > Stacks

CloudFormation

- Stacks
- Stack refactors New
- StackSets
- Exports
- Infrastructure Composer
- IaC generator
- Hooks overview
- Invocation summary
- Hooks

Stacks (2)

Delete stack | Update stack | Stack actions | Create stack

Filter status: Active

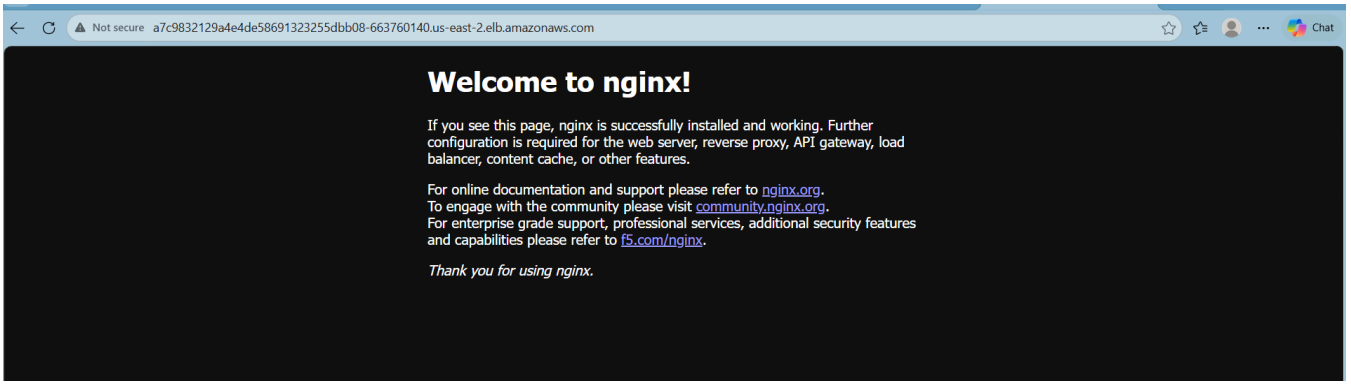
Search by stack name | View nested

Stack name	Status	Created time	Description
eksctl-demo-eks-nodegroup-demo-nodes	CREATE_COMPLETE	2026-03-15 19:40:24 UTC+0530	EKS Managed Nodes (SSH access: false) [created by eksctl]
eksctl-demo-eks-cluster	CREATE_COMPLETE	2026-03-15 19:29:42 UTC+0530	EKS cluster (dedicated VPC: true, dedicated IAM: true) [created and managed by eksctl]

Deployed Nginx

```
2026-03-15 14:13:47 [ℹ]  kubectl command should work with "/root/.kube/config", try 'kubectl get nodes'
2026-03-15 14:13:47 [✔]  EKS cluster "demo-eks" in "us-east-2" region is ready
root@ubuntu:~# kubectl get nodes
NAME                                           STATUS   ROLES    AGE     VERSION
ip-192-168-66-214.us-east-2.compute.internal   Ready    <none>   3m20s   v1.34.4-eks-f69f56f
root@ubuntu:~# kubectl create deployment nginx --image=nginx
deployment.apps/nginx created
root@ubuntu:~# kubectl get pods
NAME                     READY   STATUS    RESTARTS   AGE
nginx-66686b6766-smskw   1/1     Running   0          23s
root@ubuntu:~#
```

ubuntu | 7% | 0.68 GB / 3.78 GB | 0.01 Mb/s | 0.00 Mb/s | 21 hours | abi (x3) | /: 50% | /boot: 13%

Exposed Nginx Outside Cluster

```
root@ubuntu:~#
root@ubuntu:~# kubectl expose deployment nginx --type=LoadBalancer --port=80
service/nginx exposed
root@ubuntu:~# kubectl get svc
NAME         TYPE           CLUSTER-IP       EXTERNAL-IP                                                          PORT(S)        AGE
kubernetes   ClusterIP      10.100.0.1       <none>                                                               443/TCP        14m
nginx        LoadBalancer   10.100.107.243   a7c9832129a4e4de58691323255dbb08-663760140.us-east-2.elb.amazonaws.com   80:32143/TCP   49s
root@ubuntu:~#
```